Saad Ajmal BSEDS-23070

Waiz Jawad BSEDS-23016

Project SafeCart: Pre-Market Chemical Hazard Prediction

Objective

Can a moderator spot a chemical hazard from a listing alone before anyone buys the product? That is the question SafeCart tries to answer. Using only listing metadata, the model predicts whether a new e-commerce listing in five high-risk categories e.g (cosmetics, toys, jewellery, childcare articles and electrical appliances) is likely to have a chemical risk such as lead or mercury or phthalates. There is still a doubt in achieving perfect results but even a decent ranking would help a small moderation team decide what to check first.

Stakeholder and Users

Stakeholder: Head of Trust & Safety. Today the team mostly reacts to complaints, after the product is already sold. The goal is to move towards screening before sale.

End-Users: Listing moderators. They get a Red/Amber/Green flag the main factors behind it and similar past EU alerts with photos.

Data and Modeling

Data Source: EU Safety Gate alerts, published weekly, from January 2015 to September 2026. After dropping 12 repeated alert numbers (we kept the latest edit of each) there are 31,140 alerts, and 17,384 of them sit in our five categories.

Model Type: We will use simple baseline rules as benchmarks, logistic regression to predict chemical hazards from product details, and optionally gradient boosting to test whether a more complex model performs better. Model weights or SHAP will help explain flags to moderators. CLIP will compare product images and text, while nearest-neighbour search will find the five most similar products already flagged by the EU. A colour-coded system will assign Red, Amber, or Green ratings based on hazard scores and similarity thresholds. If needed OCR will extract product names and brands from packaging images.

Features: Product type, category, brand, country of origin and counterfeit status. The alert description is left out on purpose because it often names the hazard directly and would leak the answer.

Train/Test Split: Train on earlier years and test on the latest two. A random split would flatter us as alerts from the same period tend to look alike.

Regional Impact

As far as we know this is one of the first attempts to use Safety Gate data for machine learning based e-commerce screening. For Pakistani or any developing country marketplaces where moderation is still mostly reactive a tool built on EU evidence could fill a real gap as long as we stay honest about the domain gap above.